TEJAS PRAKASH D

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Portfolio — Google Cloud Skills Boost

PROFESSIONAL SUMMARY

AI Engineer specializing in Agentic AI, Large Language Model (LLM) applications, and MLOps, with expertise in architecting production-grade multi-agent systems, Retrieval-Augmented Generation (RAG), prompt engineering, workflow orchestration, model optimization, and AI observability. Passionate about building scalable, reliable, and secure AI platforms that transform research into production-ready intelligent systems.

EXPERIENCE

Agentic AI Engineer & MLOps Engineer June 2026 – Present
Boostt.ai Bangalore, Karnataka

- Architected and migrated AI agent workloads from AWS Lambda to **AWS ECS Fargate**, integrating **Amazon ALB** for highly available traffic routing and persistent streaming execution, achieving **95% uptime** and improved **p95/p99 latency**.
- Led and executed the AWS cloud infrastructure provisioning using **Terraform** and developed scalable **microservices**, containerized with **Docker** and deployed on **Amazon EKS** for reliable and scalable production workloads.
- Orchestrated an **AWS SageMaker** MLOps platform for broker compliance, using **Amazon S3** for centralized data storage and **AWS Glue** for automated ETL, schema discovery, and data cataloging, standardizing heterogeneous compliance data for ML workflows.
- Automated training, validation, deployment, and monitoring with **SageMaker Pipelines** and Model Registry, reducing manual ML operations by **60%**, cutting deployment time by **45%**, and supporting **10M+** compliance records.
- Integrated **Amazon Bedrock Nova Canvas** into a production GenAI workflow for high-quality image generation, using prompt-driven generation and **AWS-native orchestration** to automate enterprise content creation, reducing generation time by **70%** and increasing production throughput by **3x**.
- Pioneered a secure and scalable **Model Context Protocol (MCP)** boundary using Anthropic's **FastMCP**, decoupling AI agents from backend services through standardized tool interfaces and enabling secure, extensible, and independently scalable service integration.
- Optimized and iterated the **loop-based agent workflows** with AI guardrails, tool-call mechanism and multi-agent coordination, ensuring goal completion while improving task accuracy to approximately **95–99%** across production AI workflows.
- Implemented an event-driven messaging layer using **Amazon SQS** for asynchronous processing, buffering, and retries and **Amazon SNS** for event fan-out across distributed AI services; reduced service coupling, prevented event loss during traffic spikes, and improved platform scalability and fault tolerance.
- Built a centralized observability platform using **Amazon CloudWatch** for distributed log aggregation, error classification, RCA and **UI-driven reporting** ; integrated **LangFuse** and **Prometheus/Grafana** for LLM and infrastructure tracing, reducing manual monitoring effort by **80%**.

Applied AI Engineer Oct 2024 – Apr 2026
Tech4Biz Solutions Bangalore, Karnataka

- Architected an enterprise **AI Orchestration Platform** for reliable agentic workflow execution using prompt engineering, reflection loops, **LangGraph**, CrewAI, and AutoGen, with LangFuse observability, reducing workflow failures by **75%**.
- Built a production-grade **LLM Security Gateway** to protect enterprise AI applications through prompt injection defense, jailbreak detection, PII redaction, structured validation, and adaptive guardrails, blocking over **80%** of adversarial requests.
- Designed a scalable **Knowledge Intelligence Platform** using **LangChain-powered RAG**, hybrid retrieval, contextual reranking, and metadata-aware search to deliver grounded enterprise intelligence, improving contextual accuracy by **35%**.
- Optimized enterprise **AI Model Platforms** spanning medical vision and model serving, leveraging **U-Net**, **ViT**, **ResNet**, PEFT, LoRA, QLoRA, mixed-precision inference, intelligent batching, and CUDA optimizations, reducing inference latency by **55%** and GPU utilization by **60%**.

AI/ML Consultant (Freelance) Oct 2023 – Present
Independent Research & Consulting Bangalore, Karnataka

- Led end-to-end **AI consulting and research initiatives** for **students, startups, and PhD researchers**, architecting production-ready NLP, computer vision, generative AI, and LLM systems using **PyTorch**, **TensorFlow**, **FastAPI**, and **HuggingFace**, while mentoring **engineers** in transformers, RAG pipelines, vector search, and modern AI application development.

Software Engineering Intern June 2023 & June 2026
JPMorgan Chase & Co. Bangalore, Karnataka

- Built an event-driven **financial transaction processing service** leveraging **Spring Boot**, **Apache Kafka**, and configurable messaging pipelines to consume, deserialize, and process high-volume transaction streams, integrating external incentive services into resilient, real-time payment workflows.
- Engineered a reliable **transaction management layer** implementing atomic balance updates, relational data persistence, and scalable REST APIs using **Spring Data JPA**, **H2**, and comprehensive integration testing, ensuring end-to-end data integrity across distributed messaging and database operations.

EDUCATION

Bachelor of Technology in Information Technology Aug 2019 – Jun 2023
Presidency University, Bangalore CGPA: 8.44/10

Data Science and Analytics – IBM Certified Apprenticeship Jun 2023 – Jul 2024
Learnbay, Bangalore

TECHNICAL SKILLS

Programming Languages: Python, Java, Go, TypeScript, JavaScript, C/C++, HTML, CSS

AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, Transformers, LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, OpenCV, ONNX, PEFT, LoRA, RAG, NLP, Computer Vision, Deep Learning, Object Detection, Image Segmentation, Feature Extraction, Model Training & Evaluation

Backend & Infrastructure: FastAPI, Flask, Spring Boot, Docker, Kubernetes, Kafka, Apache Airflow, Dagster, MLflow, REST APIs, Microservices

Cloud & Databases: AWS, GCP, Azure, PostgreSQL, MongoDB, Elasticsearch, Snowflake, Terraform, Grafana, Prometheus

Tools: PowerBI, Tableau, Github, Jira, Confluence, Claude Code, Github Copilot, ChatGPT, Cursor, LabelImg (annotation)

PROJECTS

Real-Time Event Pattern Detection System | *PySpark, Kafka, Isolation Forest, LSTM Autoencoder* GitHub Link

- Architected a scalable streaming data pipeline using **PySpark** and **Apache Kafka** to ingest, parse, and process high-volume event streams concurrently in real time.
- Engineered unsupervised ML anomaly detection models (**Isolation Forest** and deep **LSTM Autoencoders**), achieving **88% precision** in identifying anomalous behavioral threats at production scale.

LLM Red-Teaming Framework | *Python, LangChain, OpenAI API, HuggingFace*

- Developed an automated security testing framework using **Python** and **LangChain** to evaluate safety boundaries and systemic vulnerabilities across enterprise LLM systems.
- Systematically executed and analyzed **500+ security attack vectors**, effectively validating model resilience against jailbreaks, complex role-play injections, and payload splitting techniques.

Voice AI Gateway | *FastAPI, WebSockets, STT/TTS, LLM Integration* GitHub Link

- Built a high-performance middleware gateway leveraging **FastAPI** and stateful **WebSockets** to enable low-latency, bidirectional real-time audio and voice streaming.
- Orchestrated an asynchronous pipeline integrating complex **Speech-to-Text (STT)**, predictive LLM logic execution, and text synthesized **Text-to-Speech (TTS)** audio loops.

CERTIFICATIONS

Python & Statistics/ML – IBM & Learnbay (2024) — AWS Cloud Practitioner – AWS (2021) — Cybersecurity – Cisco (2021)